

COMPLETE FOLDER STRUCTURE SPECIFICATION

BCA Department Website — Exhaustive Directory & File Mapping

OFFICIAL FILE MAP

This official technical document presents the 100% complete, file-by-file directory structure of the BCA Department Website repository. Every single component, context provider, API controller, route, Prisma model migration, static resource, utility function, and configuration file is explicitly listed with inline architectural annotations.

TOTAL FILES MAPPED

250+ Source Files

ARCHITECTURE PATTERN

Modular Clean Architecture

FRONTEND ECOSYSTEM

React 18 + Context API

BACKEND ECOSYSTEM

Node.js + Express + Prisma

13.1 Frontend Directory Structure (`src/components/`)

The codebase follows strict separation of concerns, dividing presentation components into specialized domain folders.

```
src/
├── components/                                # Reusable & Feature-Specific UI Components
│   ├── auth/                                # Authentication & Modal Forms
│   │   ├── AuthModal.jsx                    # Login / Signup Modal Container
│   │   ├── LoginForm.jsx                    # Password & Email Auth Form
│   │   ├── RegisterForm.jsx                # Student Account Registration Form
│   │   └── SocialLoginButtons.jsx           # OAuth Social Login Triggers
│   ├── contest/                             # Competitive Coding Arena Components
│   │   ├── ContestCard.jsx                 # Contest Overview Card with Live Timer
│   │   ├── ContestHeader.jsx               # Arena Navbar with Remaining Time Gauge
│   │   ├── ContestLeaderboard.jsx          # Real-Time ICPC Scoring Table
│   │   ├── ContestProblemList.jsx          # Problem Selection Panel
│   │   ├── ContestSubmissionHistory.jsx    # Student Verdict History List
│   │   └── ContestTimer.jsx                # Synchronized Countdown Clock
│   ├── dashboard/                           # Student Analytics & Widgets
│   │   ├── ActivityHeatmap.jsx             # Daily Coding Contribution Grid
│   │   ├── DashboardHeader.jsx             # Student Welcome & Level Banner
│   │   ├── QuickAccessGrid.jsx             # Shortcuts to Active Semester & Subjects
│   │   ├── RecentSubmissionsCard.jsx       # Recent Judge0 Verdict Stream
│   │   ├── SkillRadarChart.jsx             # Proficiency Radar across BCA Subjects
│   │   └── StreakCounterCard.jsx           # Active Solve Streak Fire Counter
│   ├── language/                            # Technology Hub Layout Sub-Components
│   └── learning/                            # Courseware & Lesson Navigation
│       ├── ChapterAccordion.jsx            # Unit Chapter Expansion Panel
│       ├── DataStructureQuestionTemplate.jsx # DS Practical Problem View
│       ├── GeneralEnglishChapterTemplate.jsx # English Subject Chapter Template
│       ├── GlobalChapterTemplate.jsx        # Universal Subject Lesson Container
│       ├── LessonCard.jsx                  # Lesson Preview Tile
│       ├── LessonContent.jsx               # Rich Text & Code Snippet Body
│       ├── LessonHeader.jsx                # Lesson Title, Marks & Duration
│       ├── LessonList.jsx                  # Sidebar Lesson Index
│       ├── LessonMetadata.jsx              # Unit Credit & Prerequisite Tags
│       ├── LessonNavigation.jsx            # Next / Previous Lesson Buttons
│       ├── LessonProgressCard.jsx          # Topic Mastery Completion Bar
│       ├── LessonSidebar.jsx               # Sticky Table of Contents
│       ├── LessonViewer.jsx                # Primary Lesson Frame Engine
│       ├── MarkCompleteButton.jsx          # Student Lesson Completion Trigger
│       ├── NotesShortcut.jsx               # Quick Jump to PDF Notes
│       ├── ProblemSolvingCChapterTemplate.jsx # C Programming Lesson Template
│       ├── SubjectNavigator.jsx            # Semester Subject Switcher
│       └── UnitAccordion.jsx                # Collapsible Unit Syllabus List
```

13.1

Frontend Directory Structure (`src/components/`, `data/`, `context/`)

```

├── navigation/                                # App Header & Navigation Elements
│   ├── AdminNavbar.jsx                       # Admin Portal Navigation Bar
│   ├── AuthNavbar.jsx                       # Logged-in User Header
│   ├── GuestNavbar.jsx                     # Public Visitor Header
│   ├── MegaMenu.jsx                         # Multi-Column Subject Dropdown Menu
│   ├── SearchModal.jsx                     # Instant Fuzzy Search Modal (Ctrl+K)
│   └── StudentNavbar.jsx                   # Student Profile & Progress Header
├── programming/                             # Coding Practice Hub UI Elements
│   ├── EmptyProgramsState.jsx              # No Search Results Placeholder
│   ├── FilterBar.jsx                       # Difficulty & Tag Filter Chips
│   ├── LoadingSkeleton.jsx                 # Animated Shimmer Placeholders
│   ├── ProgramCard.jsx                     # Executable Code Program Tile
│   ├── ProgramHeader.jsx                   # Practice Arena Banner
│   ├── ProgramList.jsx                     # Grid of 50+ Practical Code Samples
│   ├── ProgrammingStatsCard.jsx            # Solved Counter & Accuracy Metric
│   ├── ProgramSidebar.jsx                  # Topic Category Selection Drawer
│   ├── QuickToolsCard.jsx                  # Format Code & Copy Snippet Tools
│   ├── RecentProgramsCard.jsx              # Last Practiced Code Exercises
│   ├── RewardsCard.jsx                     # Earned Achievement Badges
│   ├── SearchBar.jsx                       # Live Program Keyword Input
│   └── SubjectTabs.jsx                     # Language Switcher Tabs (C, C++, Java)
├── sections/                               # Landing Page Section Components
│   ├── About.js                            # Department Vision & Overview
│   ├── CompanyLogos.js                     # Placement Recruiting Partners Grid
│   ├── Contact.js                          # Department Helpdesk Contact Form
│   ├── Faculty.js                          # BCA Department Faculty Profiles
│   ├── FAQ.js                              # Frequently Asked Questions Accordion
│   ├── Footer.js                           # Global Copyright & Sitemap Links
│   ├── Gallery.js                          # Department Labs & Events Photo Stream
│   ├── Labs.js                             # Computer Lab Infrastructure Details
│   ├── Placement.js                         # Student Campus Placement Statistics
│   ├── Projects.js                         # Student Capstone Project Showcase
│   ├── Resources.js                        # Notes & Past Papers Download Grid
│   ├── Roadmap.js                           # Semester-by-Semester Study Path
│   ├── Stats.js                            # Key Platform Metrics & Solved Counts
│   ├── TechLogos.js                        # Technology Stack Icons Carousel
│   ├── Technologies.js                     # C, Java, Python, DBMS Overview
│   ├── TechnologyLearningHub.js            # Generic Tech Hub Container
│   ├── Testimonials.js                     # Alumni Reviews & Feedback
│   └── WhyChoose.js                         # Value Proposition Highlights
├── ui/                                     # Atomic Design System Tokens & Badges
│   ├── Badge.jsx                           # Status & Tag Pills (EASY, MEDIUM)
│   ├── Breadcrumb.jsx                      # Hierarchy Navigation Trail
│   ├── Button.jsx                          # Neon Gradient Button Component
│   ├── ContentCard.jsx                     # Glassmorphism Card Container
│   ├── GlassPanel.jsx                      # Translucent Backdrop Filter Panel
│   ├── PageLoader.jsx                      # Fullscreen Chunk Suspense Spinner
│   ├── ProgressBar.jsx                     # Dynamic Animated Completion Gauge
│   ├── SectionTitle.jsx                    # Glowing Section Header Component
│   ├── StickyTabs.jsx                      # Pin-to-top Navigation Tabs
│   └── LiveCodeEditor.jsx                  # Monaco Code Compiler Workbench
├── context/                                # React Context API State Providers
│   ├── AuthContext.jsx                     # User Session, Tokens & Role State
│   ├── ContestContext.jsx                  # Live Contest Timer & Submissions
│   ├── LearningContext.jsx                 # Active Unit & Lesson State
│   ├── NotificationContext.jsx              # Toast Alerts & System Push Alerts
│   ├── ProgramContext.jsx                   # Executable Code Examples State
│   ├── ProgressContext.jsx                  # Daily Solve Streak & Solved Counts
│   ├── SettingsContext.jsx                 # User Preferences & Editor Keybindings
│   └── ThemeContext.jsx                     # Dark Purple Theme Tokens State

```

13.1

Frontend Directory Structure (`src/pages/`, `services/`, `styles/`)

```

├─ pages/                                # Page-Level Views (36+ Lazy Chunks)
├─ About.js                             # Department Overview Page
├─ Analytics.jsx                         # Comprehensive Skill Radar & Analytics
├─ CLearningHub.js                       # C Language Comprehensive Hub
├─ CodingPractice.jsx                   # Monaco Multi-Language Compiler View
├─ ContestDetails.jsx                   # Contest Rules & Registration View
├─ ContestLeaderboardPage.jsx           # Live ICPC Ranking Table View
├─ ContestList.jsx                      # Available & Upcoming Contests
├─ ContestProblemArena.jsx              # Contest Timed Coding Workbench
├─ ContestResultsPage.jsx               # Final Contest Score & Rating Changes
├─ CppLearningHub.js                   # Object Oriented C++ Hub
├─ CppQuiz.js                           # Interactive C++ Multiple Choice Quiz
├─ Dashboard.jsx                        # Main Student Dashboard Command Center
├─ DataStructureQuestionPage.jsx         # DS Exam Question & Solution Viewer
├─ DataStructureSyllabus.js             # Prescribed DS Syllabus Outline
├─ DataStructureUnit1Page.jsx           # Linear Data Structures Unit Page
├─ DSQuiz.js                            # Data Structures Interactive Quiz
├─ GeneralEnglishChapterPage.jsx        # English Chapter Reading View
├─ GeneralEnglishSubjectPage.jsx        # General English Subject Dashboard
├─ Home.js                              # Landing Page Master View
├─ JavaLearningHub.js                   # Java Enterprise Programming Hub
├─ LessonViewerPage.jsx                 # Courseware Lesson Reading View
├─ Login.jsx                            # Student Auth & Login Page
├─ MasterLanguagePage.jsx               # Universal Language Hub Router Page
├─ Profile.jsx                          # User Account Settings & Badges
├─ ProgrammingHub.jsx                  # Code Practice Library Catalog
├─ ProgramViewerPage.jsx                # Single Code Program Executable View
├─ PythonLearningHub.js                 # Python Science & Scripting Hub
├─ Roadmap.jsx                          # Semester Learning Path Map
├─ SemesterCChapterPage.jsx             # C Programming Chapter View
├─ SemesterCSubjectPage.jsx             # C Programming Subject Portal
├─ SubjectSyllabus.js                   # Universal Syllabus Viewer
├─ curriculum/                          # Curriculum Specialized Pages
├─   └─ DSNotesPage.jsx                 # Data Structures PDF & Text Notes
├─   └─ DSQuestionsListPage.jsx         # DS Exam Question Bank Catalog
├─   └─ DSQuestionViewerPage.jsx        # Single Exam Question Solution Page
├─ services/                            # Backend REST & Firebase API Integration
├─   └─ api.js                          # Central Axios Instance with Interceptors
├─   └─ contestService.js               # Contest & Leaderboard Endpoints
├─   └─ firebaseAuth.js                 # Firebase Social Auth Integration
├─   └─ searchService.js                # Client Fuzzy Search Engine
├─   └─ socketService.js                # Real-time WebSocket Client
├─   └─ auth/                           # Auth Service Module
├─     └─ authService.js                # JWT Login, Register & Token Refresh
├─ styles/                              # Modular Vanilla CSS Files
├─   └─ global.css                      # Master CSS Variables & Typography
├─   └─ tokens.css                      # Design System Color & Radius Tokens
├─   └─ Dashboard.css                   # Student Dashboard Layout Styles
├─   └─ LiveCodeEditor.css              # Monaco Code Compiler Workbench Styles
├─   └─ design-tokens/                  # JS Token Exports (Colors, Spacing)
├─ App.js                               # Master Client Router & Suspense Fallbacks
└─ index.js                             # React DOM Rendering Entry Point

```

13.2 Backend Directory Structure (`backend/src/`)

The backend architecture uses modular domain-driven TypeScript layers with Prisma ORM, Express routers, and BullMQ execution queues.

```

backend/src/
├── config/                                # Server Environment & DB Setup
│   ├── app.config.ts                     # Express App Port, CORS & Security Config
│   ├── auth.config.ts                   # JWT Secret, Expiration & Cookie Flags
│   ├── database.ts                       # Prisma Client Connection Pool Config
│   ├── logger.config.ts                 # Pino Logger Settings
│   └── queue.config.ts                  # Redis BullMQ Connection Credentials
├── middleware/                           # Global Express Request Handlers
│   ├── error-handler.middleware.ts      # Global Exception & Prisma Error Guard
│   ├── maintenance.middleware.ts        # System Emergency Maintenance Interceptor
│   └── request-context.middleware.ts     # Unique Correlation ID Injector
├── modules/                             # Domain Business Modules
│   ├── auth/                            # User Identity & Token Management
│   │   ├── auth.controller.ts           # Express Login, Register & Logout Handlers
│   │   ├── auth.routes.ts              # Auth Endpoint Definitions
│   │   ├── auth.validator.ts           # Zod Schema Request Validation
│   │   ├── middleware/auth.middleware.ts # JWT Bearer Token Verification Guard
│   │   ├── providers/bcrypt.provider.ts # Password Hashing Provider
│   │   ├── repositories/user.repository.ts # Prisma User Table Data Access
│   │   └── services/token.service.ts    # SHA-256 Token Rotation Service
│   ├── contest/                        # Timed Competitions & Scoring Engine
│   │   ├── contest.controller.ts       # Contest CRUD & Problem Fetching
│   │   ├── contest.routes.ts           # Contest API Routes
│   │   ├── gateways/contest.gateway.ts # Socket.io Live Score Publisher
│   │   ├── repositories/contest.repository.ts # Prisma Contest Table Queries
│   │   └── services/scoring.service.ts  # ICPC Time Penalty Calculation
│   ├── operations/                     # Admin Dashboard & Moderation
│   │   ├── operations.controller.ts     # System Metrics & User Ban Handlers
│   │   └── services/rbac.service.ts      # Dynamic Permission Group Validator
│   ├── problem/                        # Practice Problem & Test Case CRUD
│   │   ├── problem.controller.ts        # Question Bank Query Handlers
│   │   ├── problem.routes.ts            # Problem Endpoints
│   │   └── repositories/problem.repository.ts # Prisma Problem Data Access
│   ├── profile/                        # Student Profile & Analytics
│   │   ├── profile.controller.ts        # User Stats & Streak Fetcher
│   │   └── services/rating.service.ts    # Glicko-2 Competitive Rating Calculator
│   ├── queue/                          # BullMQ Background Job Processing
│   │   └── workers/submission.worker.ts  # Async Judge0 Payload Execution Queue
│   ├── realtime/                       # Socket.io WebSockets Layer
│   │   └── gateways/socket.gateway.ts    # Live Leaderboard Connection Gateway
│   ├── submission/                     # Code Execution Submission Pipeline
│   │   ├── submission.controller.ts     # Code Compile Request Handler
│   │   └── services/submission.service.ts # Judge0 / Piston API HTTP Dispatcher
├── repositories/                       # Base Generic Repository Class
│   └── base.repository.ts               # Type-safe Prisma Abstract DB Layer
├── app.ts                              # Express App Initialization & Middleware Stack
└── server.ts                           # HTTP & Socket.io Server Entry Point

```

13.3

Database (^prisma/), Public Assets (^public/) & Root Files

Overview of relational database schemas, PDF courseware assets, and root environment configuration files.

```
backend/prisma/                # Database Schema & Migrations
├─ migrations/                 # Prisma Migration History
│   └─ 20260717023256_init_apex_database_foundation/
│       └─ migration.sql       # Base User, Role & Session Tables SQL
│   └─ 20260717040318_configure_refresh_token_family/
│       └─ migration.sql       # Refresh Token Rotation Tables SQL
│   └─ 20260717041805_configure_problems_and_testcases/
│       └─ migration.sql       # Problem, TestCase & Submission Tables SQL
│   └─ 20260719050057_sync_contest_schema/
│       └─ migration.sql       # Contest, Leaderboard & Score Tables SQL
├─ schema.prisma               # Master Prisma Database Schema (30+ Models)
└─ seed.ts                     # Sample Student & Problem Seeding Script

public/                         # Static Assets & Course Documents
├─ notes/                      # Downloadable Unit Notes PDFs
│   └─ semester2/
│       └─ DS_Notes.pdf        # Data Structures Comprehensive Notes
├─ papers/                     # University Past Examination Papers
│   └─ cpp/
│       ├── cpp-paper-2025.pdf  # Summer 2025 C++ Exam Paper
│       └─ cpp-paper-2026.pdf  # Winter 2026 C++ Exam Paper
│   └─ semester2/
│       └─ OOP_Java_Winter_2025.pdf # Winter 2025 Java Exam Paper
├─ syllabus/                   # Official University Course Outlines
│   ├── ds-syllabus.pdf         # Data Structures Syllabus PDF
│   └─ os-syllabus.pdf         # Operating Systems Syllabus PDF
├─ favicon.ico                 # Department Platform Browser Icon
├─ index.html                  # Single Page Application Main HTML Host
└─ pdf.worker.min.js           # PDF.js High Performance Rendering Worker

Root Directory /               # Platform Build & Deployment Configs
├─ .env.example                # Template for API & DB Credentials
├─ docker-compose.yml          # Local PostgreSQL & Redis Container Config
├─ package.json                # Frontend NPM Dependencies & Build Scripts
├─ README.md                   # Project Setup & Developer Manual
├─ start-project.bat            # One-Click Windows Development Startup Script
└─ vercel.json                  # Production Edge CDN Route Rewrites
```

END OF COMPLETE FOLDER STRUCTURE SPECIFICATION

BCA DEPARTMENT WEBSITE — OFFICIAL CODEBASE DIRECTORY MAP